

Solar-Energy Systems

An alternate and renewable source of power generation has become inevitable to meet the ever-increasing power demand. In this situation, solar-power generation has become the need of the day

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'Solar' is the Latin word for 'sun.' Solar energy from sun gives life to earth and its creation. Besides this, solar energy can also be successfully utilised for various applications such as cooking, heating, drying, lighting, power generation, communication, space technology, etc. Our country is blessed with abundant sunlight, which is a very large resource for development of solar photovoltaics (PV) power. There is no negative impact on the environment with utilisation of this power and it is freely available. Also, it is the safest renewable energy resource available in the world.

Solar cells, commonly called as PV cells, convert sunlight directly into electricity. The name PV is derived from the process of converting light (photons) into electricity (voltage).

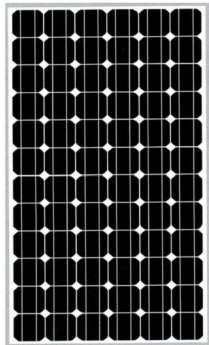


Fig. 1: Solar panel

This is called the 'PV effect.' The PV effect was discovered in 1954 by the scientists at Bell Telephone Company. They discovered that silicon (an element found in sand) creates an electric charge when it is exposed to sunlight. Since then, these solar cells have been used to operate smaller devices like calculators, watches, lights, etc and also large aircraft and space satellites. In the present day, thousands and thousands of people power their houses, business establishments, manufacturing units, etc with solar PV systems.

In a solar-energy system, the most

important component is a solar or PV cell. This is a solid-state device with a large light-sensitive area that converts the light energy from the sun directly into electricity by the PV effect. To increase the power, a number of individual PV cells are interconnected in a weatherproof sealed package called a PV module or a solar panel. Solar cells are very brittle and require great care and protection whilst handling. To protect them from damage, they are hermetically sealed between top layer of glass or transparent plastic and the bottom layer is protected with plastic sheet or a combination of metal and plastic. To increase the strength of the solar panel, the outer edge assembly is kept in a metal frame all the time.

A junction box is fixed at the bottom side of solar panel to interconnect and take the output voltage. A mounting frame made in metal is used to assemble these solar panels which are connected in parallel or a series combination as per requirement of the voltage and current.

Solar-home system

A solar-home system consists of a solar PV module, a battery bank and a power-conditioning unit (PCU). The output voltage from the PCU can

operate light, fan, TV, computer, home appliances, etc depending on the capacity of the system.

Solar PV module generates DC voltage in proportion to solar panel size and the available sunlight. The power generated in the PV module is sent to the charge control



Fig. 2: Solar-panel assembly



Fig. 3: Solar power plant